

QuAD: Deep-Learning Assisted Qualitative Data Analysis with Affinity Diagrams

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ABSTRACT

Affinity diagramming is an effective and efficient method for forming nuanced interpretations of wide-ranging, unstructured qualitative data; however, this method does not scale well to large data sets. We propose a novel affinity diagramming system, called Qualitative Affinity Diagrammer (*QuAD*) that leverages computer-generated suggestions using deep learning to address the scalability of the diagramming process. *QuAD* features automatic grouping suggestions to jump-start the affinity diagramming process and provides grouping suggestions throughout the diagramming process to reduce sifting through notes. In this paper, we present a prototype of *QuAD* that uses Bidirectional Encoder Representations from Transformers (BERT) and Girvan-Newman to generate grouping suggestions. This work is the first step towards creating a powerful tool for assisting in the analysis of large qualitative data sets in a variety of contexts, including human-computer interaction.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Open source software.

KEYWORDS

Qualitative data, affinity diagramming, natural language processing, deep learning

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1 INTRODUCTION

Affinity diagramming is a collaborative methodology that allows researchers to build a shared interpretation of varied, unstructured and seemingly disparate qualitative data and synthesize overarching themes [19]. Affinity diagrams are constructed by first breaking the data down into individual “notes” (such a short interview quote) that are subsequently grouped with related notes to reveal themes to label. Labeled themes are iteratively grouped until an overarching story describing the data emerges [2]. This method of qualitative data analysis is used in a variety of fields. For example, affinity diagramming is a common technique in Contextual Design and Human Computer Interaction (HCI) research to analyze interview and observational data [19]. Affinity diagramming can also be used to help businesses better understand their users’ needs. For example, Judge and McCrickard [20] describe how software developers, usability engineers, and managers at a company could use affinity diagramming to assist in the development of a new internal time and billing system for their company.

Although affinity diagramming is typically more efficient for large data sets than alternatives such as qualitative coding, the

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diagramming process is inefficient and overwhelming as current methods are optimized for small data sets [2]. Once the collection of notes is too large, the diagrammers often have difficulty keeping track of all the notes and how they fit into the diagram. Due to these limitations, researchers with data that is well suited to being diagrammed (qualitative, unstructured, varied) may avoid affinity diagramming if their data set is large (e.g., ethnographic interviews with many participants, such as O'Carroll et al.'s interviews with repeal activists [26]), causing them to miss out on the nuanced data interpretations affinity diagramming can provide.

In this paper, we propose a novel affinity diagramming system, Qualitative Affinity Diagrammer (QuAD), that leverages deep learning to provide grouping suggestions that address the issues of the scalability and efficiency in affinity diagramming. This makes affinity diagramming of large qualitative data sets more practical. The contributions of this paper include 1) a method combining deep learning and data clustering for automatically grouping qualitative data to be fed into an affinity diagram; and 2) QuAD: an open-source system for facilitating affinity diagramming.

2 RELATED WORK

2.1 Affinity Diagramming

Affinity diagramming, otherwise known as the KJ method, is a method of qualitative data analysis [22, 28]. This method begins by separating the qualitative data into individual notes [2]. Then, the notes are grouped by affinity or similarity, and the groups are labeled with a statement that communicates the groups' common theme. These labels are then themselves grouped, and these higher level groups are given labels, revealing overarching themes and concepts that provide new insights into the initial qualitative data. This process repeats iteratively to form the affinity. After the affinity diagram has been made it can be walked, which involves reading and annotating the diagram with thoughts or questions [3]. For example, if an affinity diagram contained data from a user study for a product, someone walking the wall could place a potential solution to a user difficulty with the product on the wall, next to the part of the diagram illustrating the problem.

A commonly used version of the affinity diagramming process was developed by Beyer and Holzblatt [2]. This method is designed to handle small data sets and focuses on encouraging collaboration. As a result, Beyer and Holzblatt recommend assigning each diagrammer 100 notes and striving to build the entire diagram in a single day so that the process does not become exhausting. However, a key drawback to this method is the potential to overwhelm users if there are too many notes or not enough diagrammers. QuAD aims to address this issue by streamlining the grouping process so that diagrammers can process more notes individually.

2.2 Computer Assisted Affinity Diagramming

There has been substantial prior work on the ways technology can assist in the affinity diagramming process. A variety of formats and functionalities have been explored to provide this assistance. For example, the ability to manipulate notes as a group has been explored in prior work [11, 32], which allows users to efficiently move notes around as they are organizing their ideas and building the diagram. As a result, we designed QuAD to expand on prior

work by designing our grouping mechanism in a way to minimize clutter during a large scale grouping process.

Previous research has also explored virtual [11, 20, 32, 33] vs physical [15, 19, 30] formats for affinity diagramming. Unlike physical systems that require large amounts of space and can result in notes falling off the diagram, virtual systems that feature virtual spaces are relatively space-efficient and stable. In addition, using individual devices and a shared vertical screen to work on a shared diagram allows users to focus on tasks individually or in small groups on personal devices while seeing the big picture on the shared screen [20, 32, 33]. In contrast, systems that utilize a shared horizontal space limit the number of collaborators who can see the system at the same time [11]. For this reason, QuAD is designed to use both small individual tablets and a large vertical virtual space for affinity diagramming.

However, there are still unexplored opportunities to utilize technology to improve the diagramming process. Notably, no prior work has investigated suggesting groupings for affinity diagramming based on the qualitative data (the text of the notes and labels). QuAD is the first step in filling this gap in the research by using the semantic meanings of the notes and labels to suggest groupings for affinity diagramming, thereby leveraging the qualitative data to improve the efficiency and scalability of affinity diagramming.

2.3 Computer Provided Inspiration, Similarities, and Sentiment Analysis

Previous research has explored how to find semantic relationships between pieces of text and how to utilize these relationships in software, e.g., software to aid in inspiration for brainstorming [1, 29] and sentiment analysis [6, 10, 14, 16, 21, 23, 34]. For example, Felix et al. developed a system to assist with document labelling that provides suggestions for labels, as well as suggestions for the criteria for a document being placed under a label [9]. These features function by utilizing the semantic meanings of the texts being labeled. It is important to note that the use of semantic meanings and word associations to suggest groupings and labeling in affinity diagramming has yet to be explored. Thus, QuAD will be the first of its kind to utilize semantic meanings and word associations for affinity diagramming.

3 SYSTEM

Based on gaps in the prior work, we seek to determine how the affinity diagramming process can be made more efficient, particularly for large affinity diagrams. QuAD is the first step in creating an affinity diagramming tool that improves qualitative research, allowing for efficient and scalable affinity diagrams. QuAD is a web-based application powered by Google Cloud [17] and designed with the goals of making affinity diagramming 1) more efficient and manageable and 2) feasible for data sets that would otherwise be overwhelmingly large to diagram. QuAD is designed not to replace human input in affinity diagramming, but instead to assist people in diagramming and facilitate their diagramming process. The design seeks to remove the typical challenges of diagramming large data sets through the grouping and labeling suggestions, as well as through a space-saving and clutter-combating workspace design. QuAD consists of the following key components, detailed

in the subsections below: 1) physical hardware, 2) diagram building mechanisms, and 3) automatic group suggestions.

3.1 Physical Setup

QuAD features a group workspace, displayed on a large vertical screen as shown in Figure 2, that can be edited by each user from their personal tablet. The user has a personal dock to store notes and groups on a small tablet, as shown in Figure 1. By combining both the use of a vertical screen and individual tablets, QuAD aligns with the traditional affinity diagram process of building iteratively and in collaboration with other participants.

The vertical screen workspace display is utilized during the diagram building portion of the process in order to give users a view of the overall workspace and to encourage collaboration (See Figure 2). The vertical screen presents a view of the workspace that is synced with the tablets to accurately reflect the work being done. On the vertical screen there is a colored rectangle for each user, labeled with the user's name, that shows what portion of the diagram the user is currently engaged with on the tablet. By being able to see what sections of the diagram other users are engaged with, diagrammers can see who they could be collaborating with on a particular section, as well as what sections were not yet being attended to. Users are able to navigate across the workspace displayed on the vertical screen by using a two finger drag to pan across the diagram on their tablet.

Although users interact with QuAD primarily from their tablets, users can choose to view the diagram in "walk the wall" mode on the vertical screen. This view facilitates the affinity diagramming practice of viewing the diagram from a zoomed out position to encourage reflection. While in walk the wall mode, there is a comment button on the tablet workspace that allows diagrammers to add a comment anywhere on the workspace. With the vertical screen, diagrammers can view and discuss the comments made by all parties. Comments can also be removed on the tablet screen.

3.2 Diagram Building

QuAD's primary purpose is to facilitate affinity diagramming by assisting users in manipulating notes, labels, and groups as they collaborate in real time. On their tablets, each user is able to drag notes, using a long press and drag, between the group shared workspace and their personal dock, as shown in Figure 1. The user can also drag notes on top of each other to form groups of notes. A label card is made for the group at this time. These labels are editable and can be grouped, creating higher level labels. Once the group is created, a new note can then be dragged on top of the notes in the group to add the new note to the group. Notes can also be moved around inside of the group, adjusting the order of the notes, and a note can be dragged out of the group. In order to move the entire group, the user drags around the label card. To facilitate grouping labels without having a cluttered screen, users are able to collapse anything under a label, to hide it from view. Users expand and collapse groupings by tapping a tab connected to the label.

Each diagrammer has a set of notes in their dock. The dock is located at the bottom of the screen and is navigated by horizontal swiping or pressing either the left or right directional arrow. The notes that have yet to be added to anyone's dock or to the workspace

are stored in the deck. Users can request notes from the deck, which are then added to the user's dock, when they need more notes to diagram.

3.3 Automatic Group Suggestions

A key component of QuAD is its ability to use the semantic meanings of the notes to provide system generated groupings of similar notes in the deck and suggest new members of existing groups. These suggestions are presented to users to be accepted, edited, or declined, to keep the human in the loop. By showing these grouping to the users to accept, edit, or decline, rather than automatically adding the groups to the deck, the system assures the users to stay in control of the affinity diagramming process. Before deciding to accept or decline a group, the user can also edit the labels of the groups and remove singular notes from a group. Automatic group suggestions are typically used after a large amount of data is uploaded to the diagram but can be used at any time during the diagramming process. QuAD can also return a list of notes that are similar to an selected note. The user is then presented with the most semantically similar notes ordered by similarity.

Automatic group suggestions are formed using a three-step process utilizing natural language processing methods and clustering algorithms. Although many combinations of algorithms were explored during the development process (e.g., Bidirectional Encoder Representations from Transformers (BERT) [7], DistilBERT [31], BIRCH [36], Louvain [4], K-means [18], Mean Shift Clustering [35], Hierarchical Agglomerative Clustering [25], Girvan-Newman [12], and Greedy Modularity [5]), evaluation by a small team of affinity diagramming experts identified that the combination of BERT [7] and Girvan-Newman [12] consistently formed the most useful groups as determined by the number of groups formed and the quality of the groups. Thus, the current iteration of QuAD utilizes these algorithms in the following manner.

3.3.1 Determining Semantic Similarity using BERT. In order to group the notes using BERT, QuAD first determines the semantic similarity between any two notes by first vectorizing the notes and then computing the cosine similarity between them to get a "similarity score" [7, 27]. Specifically QuAD utilizes sentence_transformers package to first use BERT to create a model of the notes, and then from that model calculate the cosine similarity between the notes. QuAD is currently configured to retrieve up to 100 of the most similar notes for each note as pairings and up to 500,000 pairings total. At the end of this process QuAD has a list of pairings of notes and their similarity scores.

3.3.2 Forming Groups using Girvan-Newman. Next QuAD forms groups of notes to provide as suggestions. QuAD first forms a graph where each note is a node and every note pairing in the pairing list that has a similarity score of 0.85 or higher is an edge. Next, QuAD uses the Girvan-Newman algorithm, which is designed to detect when nodes in a graph are densely connected to one another (i.e., grouped) by removing edges from the original graph [12]. QuAD utilizes the Python package NetworkX [13] to run Girvan-Newman. The Girvan-Newman algorithm is run for 4 iterations and the resulting communities of notes become QuAD's grouping suggestions. The parameters used for BERT and Girvan-Newman

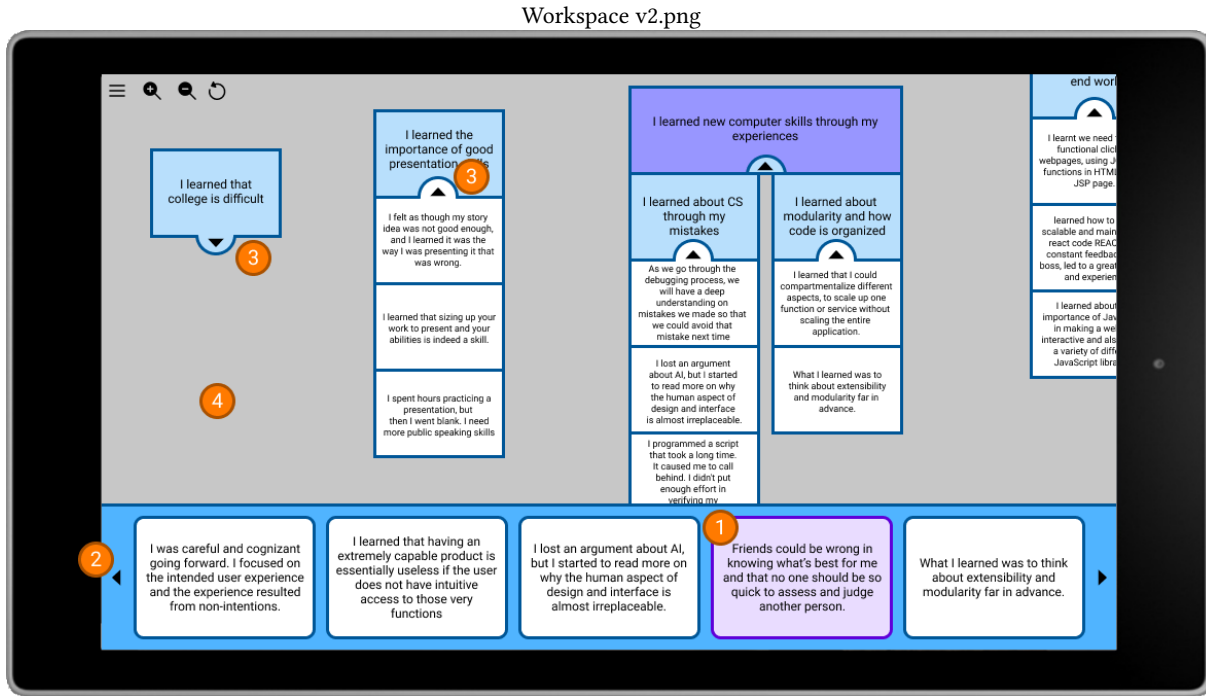


Figure 1: QuAD allows users to add notes to a shared affinity diagram from their tablet through simple interactions. (1) The user can select notes to be added to the diagram with a long press, and then drag the note to the diagram. (2) Notes in the data set that have yet to be added are presented in a dock that can be navigated via horizontal swipes or tapping one of two directional arrows. (3) Notes that have been organized under labels can be hidden or revealed by tapping a tab connected to the label. (4) The user can gain access to the parts of the diagram not shown on the tablet by panning using two finger drag.

were determined after pilot studies with a small team of affinity diagramming experts.

4 EXAMPLE: AUTOMATICALLY GROUPING NOTES FROM A CS CLASS SURVEY

To illustrate QuAD’s effectiveness, we present the results of running the automatic grouping algorithm on a data set of 431 notes from a data set of upper-level computer science student responses to an survey asking “Tell us about a time that you were wrong about something and learned from it.” Running QuAD’s automatic grouping process took 34.02 seconds and resulted in 15 automatic groups. Of these groups, nine were of a high enough quality that they are likely to be accepted by a user and included in an affinity diagram; three samples of high quality groups are shown in Table 1. In this example, quality is determined by a small group of affinity diagramming experts. Group 1, for example, is considered to be high quality because each note is primarily concerned with working in a team. Similarly, Group 2 focuses on deciding that a finance major would be the wrong choice for a student, and Group 3 illustrates situations where a simple solution can surprisingly perform better than expected.

One of the generated groups, shown in Table 2 would likely be edited by the user before inclusion in the affinity diagram. This group consists of 4 notes, three of which reflect similar themes of the importance of not pushing ahead with a project without

stopping to ask questions or ask for help. The fourth note is unrelated and instead focuses on the theme of biases and accessibility in technology. In addition, five groups were likely to be declined by a QuAD user; two examples of which (Groups 5 and 6) are shown in Table 3. These groups are formed from unrelated notes. From this, we can see that while QuAD does provide useful automatic groups, it is still necessary to keep the human in the loop of the diagramming process.

5 FUTURE WORK

As automatically suggesting groups is one of the key novel features of QuAD, future work will involve pilot evaluations to determine its effectiveness in increasing the scalability of affinity diagramming. During these pilot studies, affinity diagramming experts recruited from a class that teaches affinity diagramming at a local college will be asked to create two affinity diagrams using 200 notes randomly selected from two published datasets [8, 24] with and without QuAD’s pre-grouping feature. The conditions for these pilot studies will be counterbalanced to minimize ordering affects and will be completed on separate days to reduce fatigue effects. Users will be asked to think aloud as they build the affinity diagram and participate in a short semi-structured interview to provide qualitative insight into the affinity diagramming process both with and without QuAD’s pre-grouping feature. In addition, quantitative data

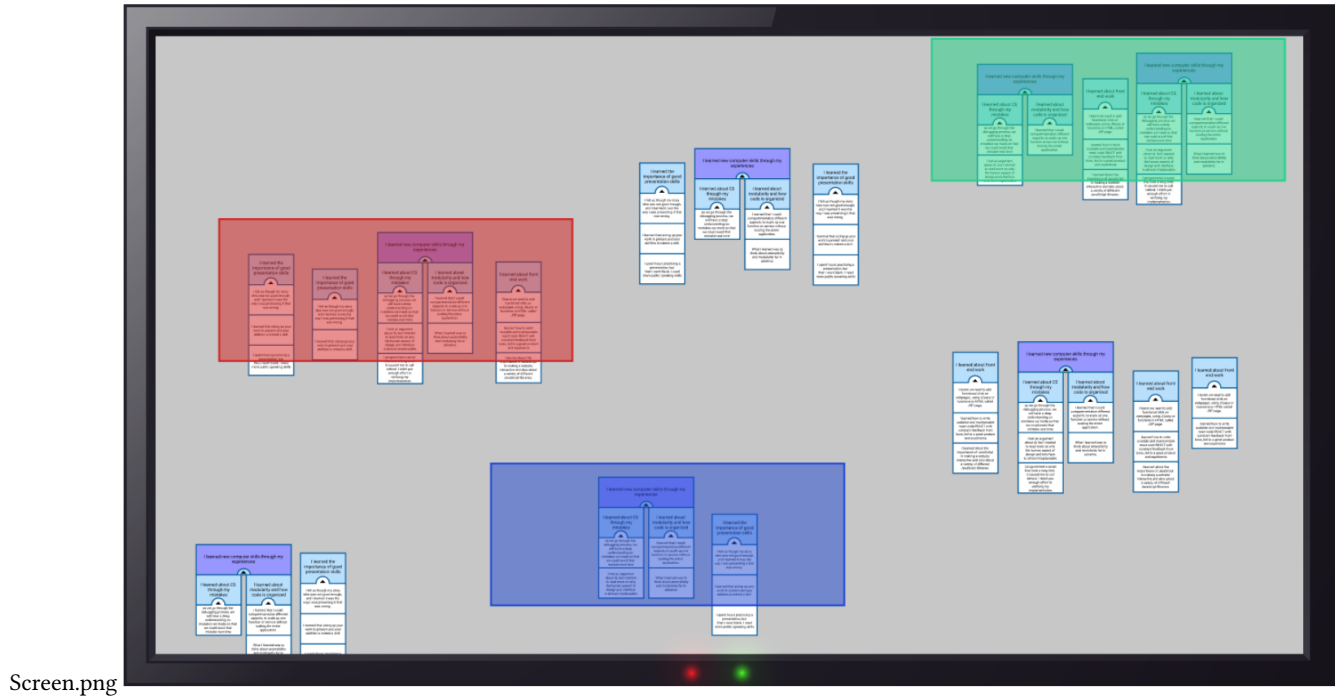


Figure 2: A large vertical display shows the entire affinity diagram with colored rectangles indicating collaborators are working in real time.

Table 1: Example automatic groups that are likely to be accepted by a QuAD user.

Group & Label	Note
Group 1: I learned the importance of teamwork	"I was in a group with people I didn't know I thought I would have to do the work, I was wrong, and my team was responsible, and hardworking."
	"I made the mistake of trying to direct every aspect of the group presentation. This showed me the importance of working in a team."
Group 2: I learned that Finance is not for me	"I thought Finance is best for me, but I was wrong because it was not for me after I had experience as an investment banker."
	"I thought I wanted to do finance, but I was wrong."
Group 3: I learned that a simple solution can surprisingly perform better than expected	"I learned that the simplest can sometimes beat the most innovative. I strive to build solutions that are better suited for the end user."
	"I learned that there are scenarios where some apparently less efficient solutions are actually more efficient and should be kept in mind."

Table 2: Example automatic group that are likely to be edited then accepted by a QuAD user.

Group & Label	Note
Group 4: I learned the importance of not pushing ahead with a project without stopping to ask for help	"I misunderstood an aspect of a project, making the final product not what was expected. I learned my lesson regarding asking questions."
	"I was asked to develop this algorithm. I chose to handle it alone, a bad mistake. I asked the PI and learned a better way to approach the task."
	"I was wrong about the scale of the group project and acquired the wrong materials. I learned to communicate & know project specifications."
	"I thought we fixed the accessibility issues, but I was wrong. I reflected on my biases, and this influenced how I think about other projects."

in the form of a survey will be gathered after each condition is completed.

Future iterations of QuAD will include additional features for improving the scalability of the affinity diagramming process. For example, QuAD's next iteration will include functionality for suggesting labels based on the semantic meaning of what is contained

Table 3: Example automatic groups that are likely to be rejected by a QuAD user.

Group	Note
5	"I learned that just because I am right doesn't mean someone else is wrong."
	"The lesson I learned is that if a thing is easy, it doesn't mean details can be ignored."
6	"I learned that sizing up your work to present and your abilities is indeed a skill."
	"I learned the value of engaging folks with institutional knowledge."

under the labels. The labeling suggestions feature is intended to speed up the labeling process and assist novice diagrammers in creating labels that capture the ideas of the group being labelled. Users will also be able to pin locations on the workspace using their tablets, and jump to these pinned locations while they are working in order to help users navigate large diagrams. Finally, users will be able to search for notes by keyword, reducing the time spent looking for specific notes.

6 CONCLUSION

In this paper, we detailed a prototype of QuAD, a system that utilizes computer generated grouping and labelling suggestions, as well as a personal tablet and shared vertical screen layout, in order to address limitations on scalability of affinity diagramming. QuAD utilizes the semantic meaning of the qualitative data in order to provide the labeling and grouping suggestions. These suggestions, both in the pre-grouping stage and the diagramming stage, help users more efficiently diagram large amounts of data. This work is the first step towards creating a powerful tool for assisting in the analysis of large qualitative data sets in a variety of contexts, including human-computer interaction.

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